

Predictive Analytics of Climate Stress Impact on the Italian Mediterranean Buffalo

Physiological and Productive Responses

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**Politecnico
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Department of Control and Computer Engineering

Dipartimento di Automatica e Informatica

Dataset Overview & Base Variables



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Dataset Overview & Base Variables

Animal Data

Total Number of Data : 2,593,267

Null values : 1873 in DTB

error values : 118,300 in All Variables

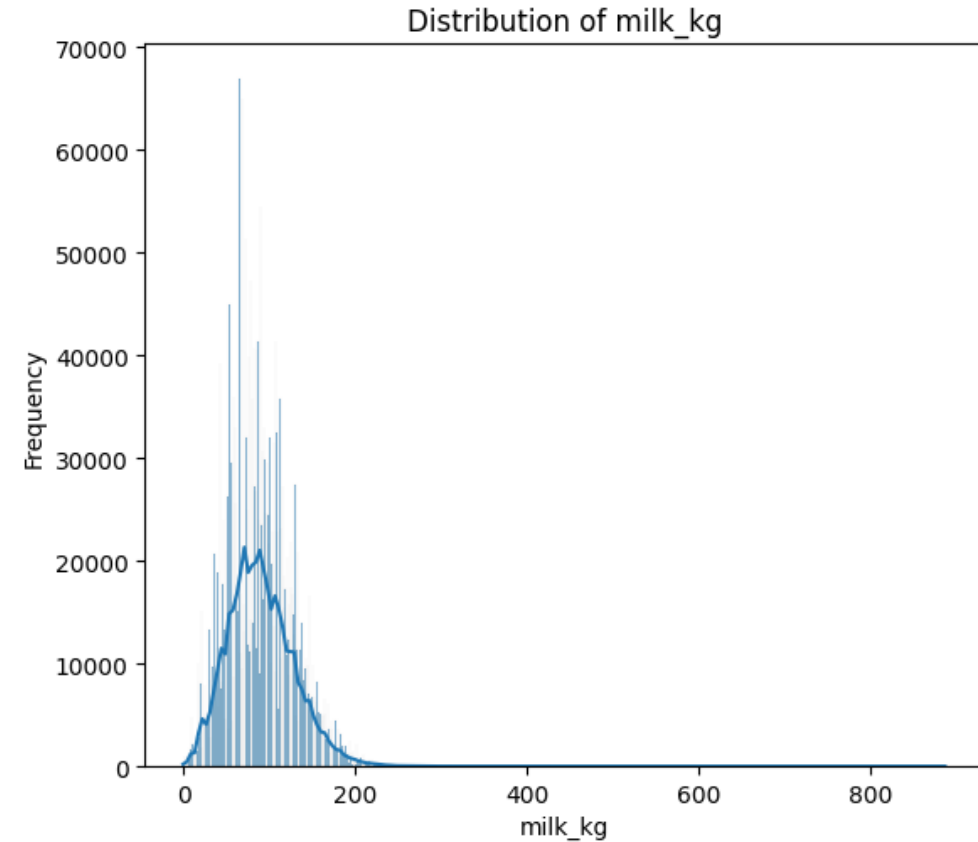
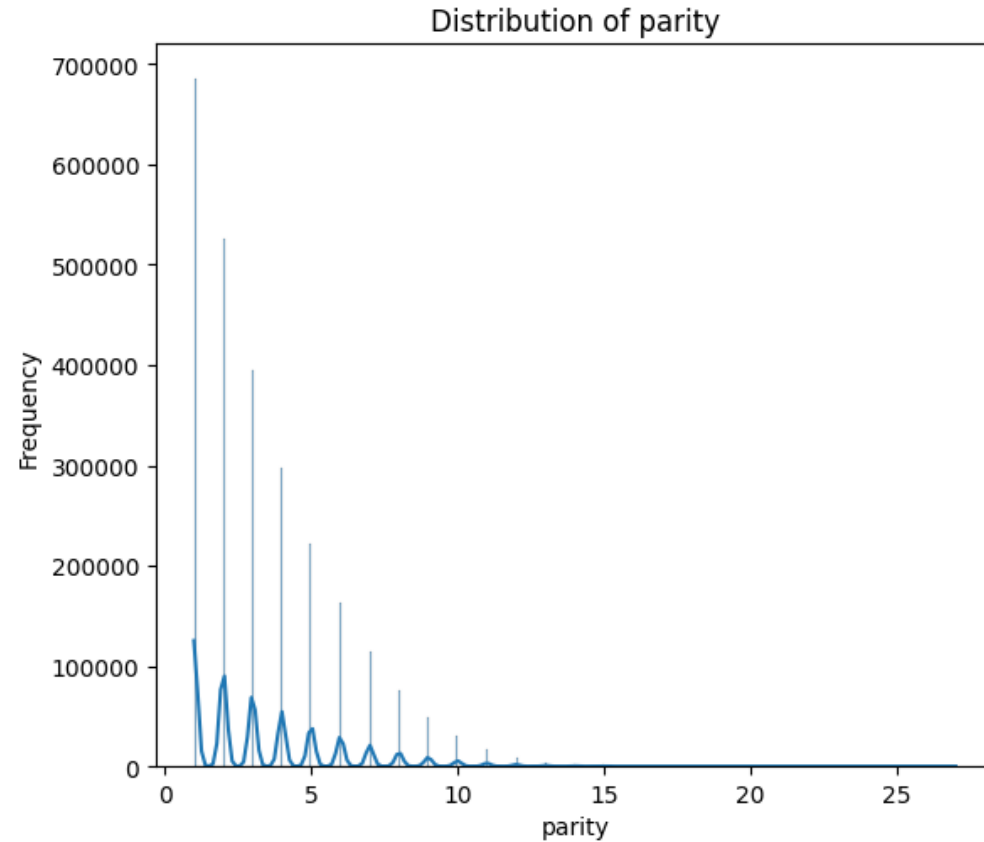
	parity	milk_kg	fat_p	protein_p	cells	NM
mean	3	90	779	457	281	2
std	2	38	239	94	798	0
min	1	0	-387	-237	0	1
max	27	886	7943	8557	98147	3

Descriptive statistics for
numerical columns

Variables	Meaning
Farm_code	Identifier of FARM
Animal_ID	Identifier Animal
DTB	Date of birth of the animal
DTC	Date of calving which provided the lactation
DTT	Date of the test-date/milk quality control
Parity	Parity order
Milk_kg	Quantitiy of milk produced per day in kg during the test
Fat_p	Percentage of Fat in the milk during the test-day
Protein_p	Percentage of protein in the milk during the test-day
cells	N. of cells per ml during the test-day
NM	N. of milking event per day

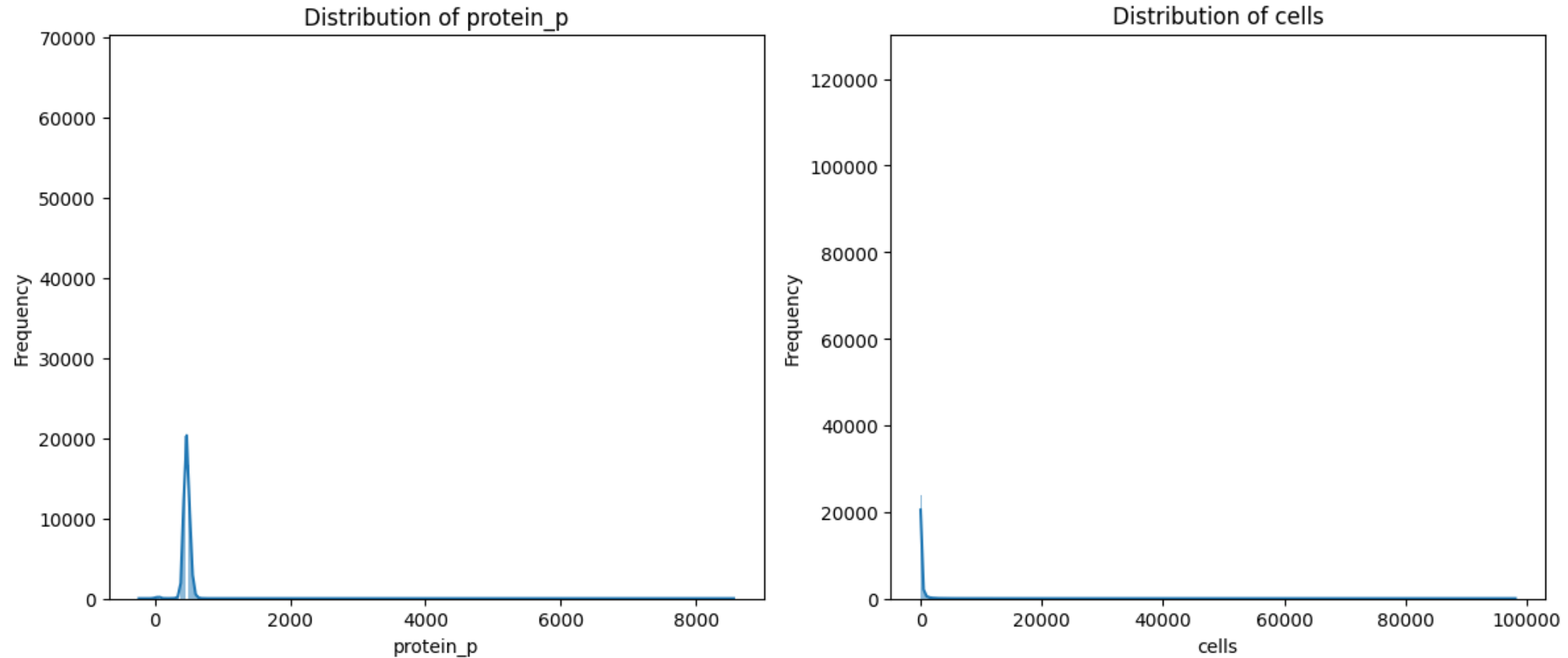
Dataset Overview & Base Variables

data visualizations for numerical columns. [Unprocessed]



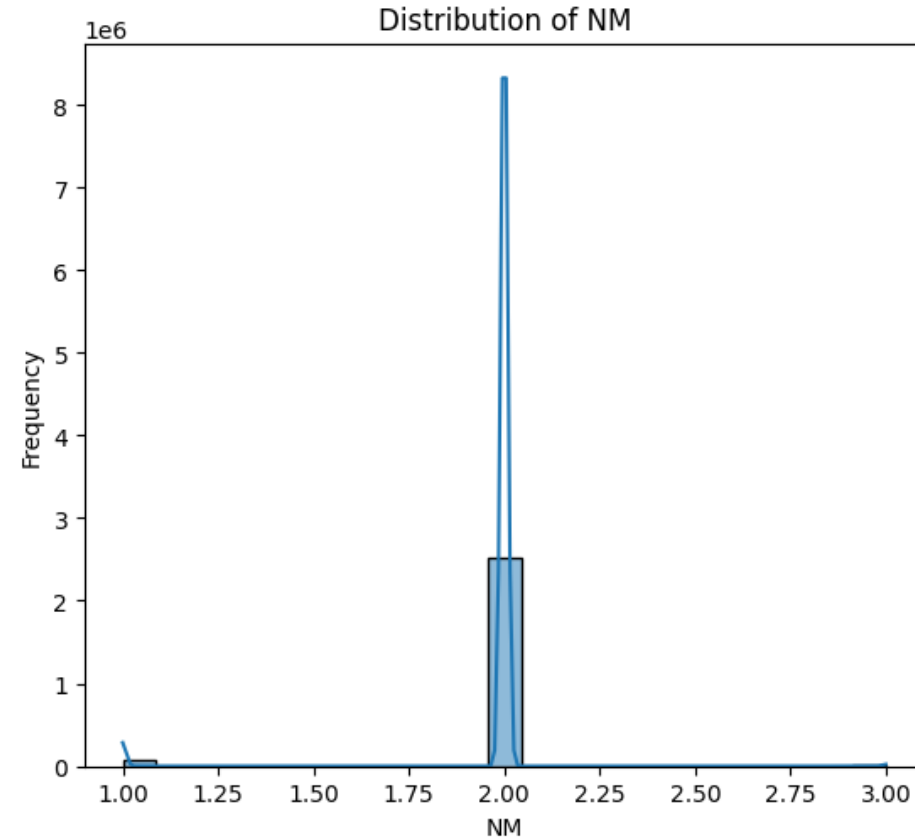
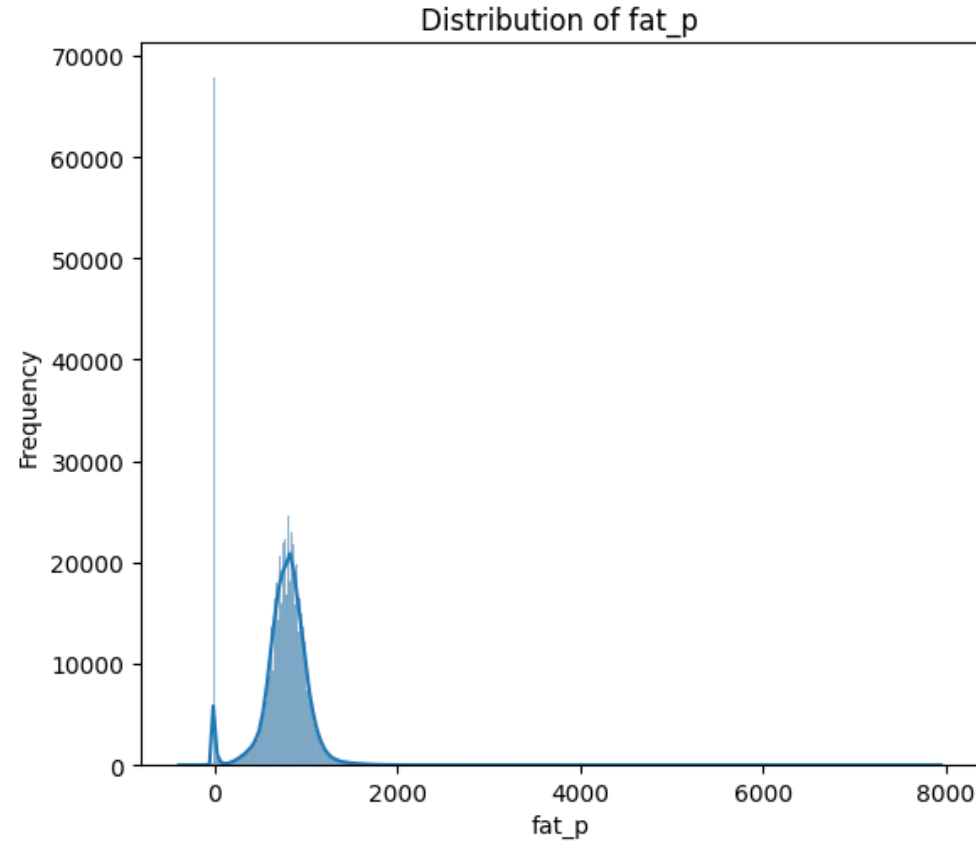
Dataset Overview & Base Variables

data visualizations for numerical columns. [Unprocessed]



Dataset Overview & Base Variables

data visualizations for numerical columns. [Unprocessed]



Dataset Overview & Base Variables

Farm Data

Total Number of Data : 329

Null values : 1 in Provinc



Variables	Meaning
Farm name	Primary name of the farm
Commercial name	Registered business or commercial name
Farm_code	Unique numerical identifier for the farm
ASL_code	Local Health Authority (Azienda Sanitaria Locale) code
Adress	Physical street address of the farm
City	City where the farm is located
Postal_code	ZIP or Postal code
Provinc	Province abbreviation (e.g., CE for Caserta)
Region	Italian region (e.g., Campania, Toscana)
Latitude	Geographic latitude coordinate
Longitude	Geographic longitude coordinate

Calculated Variables (FeatureEngineering)



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Feature Engineering

- **Scaled variables**

$$\text{milk}_{final} = \frac{\text{milk}}{10}$$

$$\text{fat}_{final} = \frac{\text{fat}}{100}$$

$$\text{protein}_{final} = \frac{\text{protein}}{100}$$

- **New Variables**

Age at First Calving (AFC)

Date of Calving

Date of Birth

$$AFC = DTC - DTB$$


Days in Milk (DIM)

Date of Test

$$DIM = DTT - DTC$$


Feature Engineering

- **New Variables**

Energy-Corrected Milk (ECM) Buffalo-specific formula:

$$ECM = ([((Fat \times 10) - 40) + ((Protein \times 10) - 31)] \times 0.01155 + 1) \times milk$$

Somatic Cell Score (SCS) Conversion to linear score:

$$SCS = \log_2\left(\frac{Cells}{100}\right) + 3$$

- **Component Thresholds**

$$milk = \begin{cases} \text{remove} & \text{if } milk \leq 0.40 \text{ or } milk \geq 26.50 \\ milk & \text{otherwise} \end{cases}$$

$$fat = \begin{cases} \text{remove} & \text{if } fat \leq 1.60 \text{ or } fat \geq 15.05 \\ fat & \text{otherwise} \end{cases}$$

$$protein = \begin{cases} \text{remove} & \text{if } protein \leq 2.67 \text{ or } protein \geq 6.68 \\ protein & \text{otherwise} \end{cases}$$

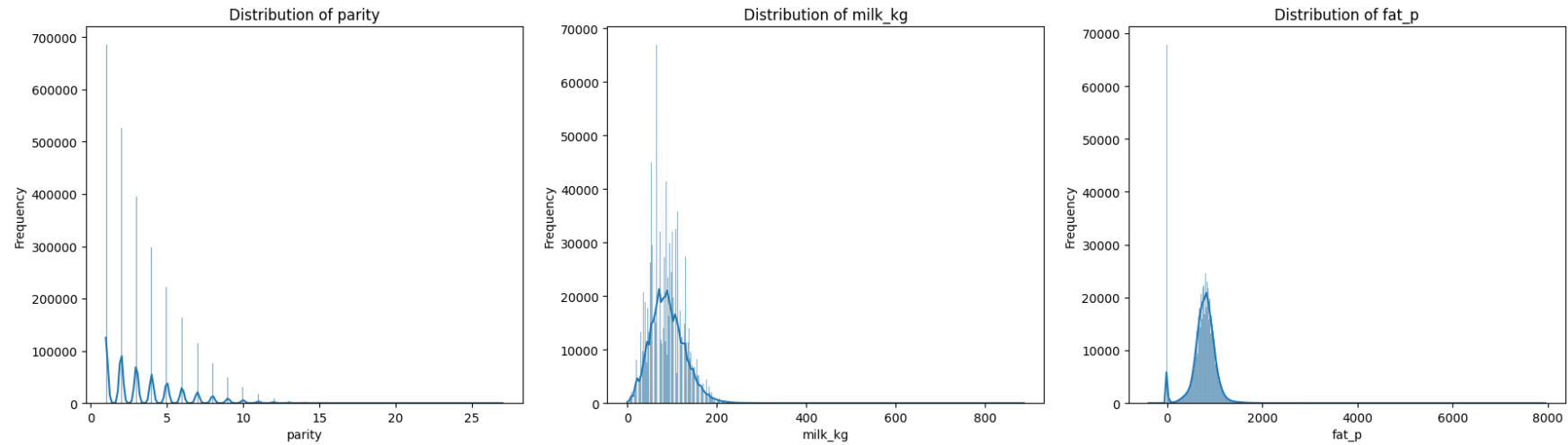
$$SCS = \begin{cases} \text{remove} & \text{if } SCS \leq -2.06 \text{ or } SCS \geq 10.73 \\ SCS & \text{otherwise} \end{cases}$$

$$ECM = \begin{cases} \text{remove} & \text{if } (milk, fat, \text{ or } protein) = 0 \\ \text{remove} & \text{if } ECM \geq 30 \\ ECM & \text{otherwise} \end{cases}$$

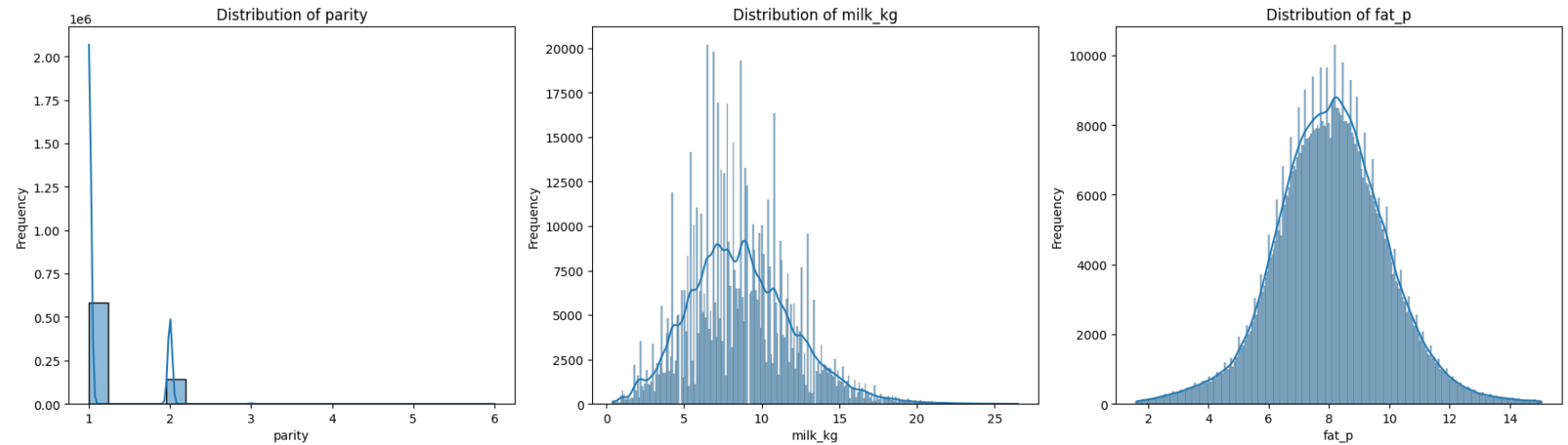
$$720 \leq AFC \leq 1440$$

Dataset Overview & Base Variables

Base Data

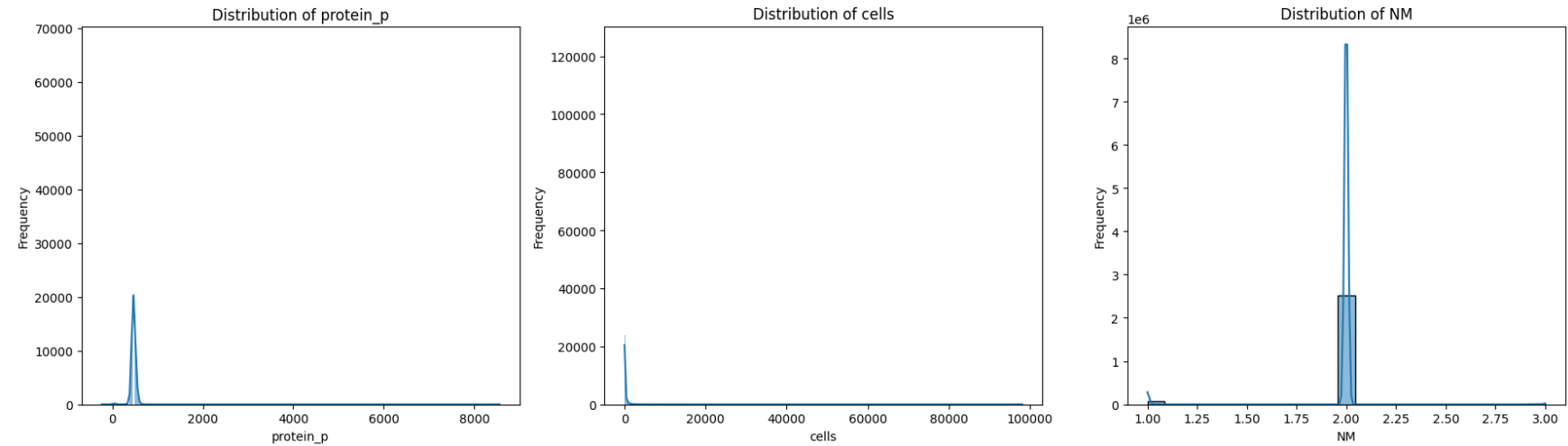


**After
Removing
outliers**

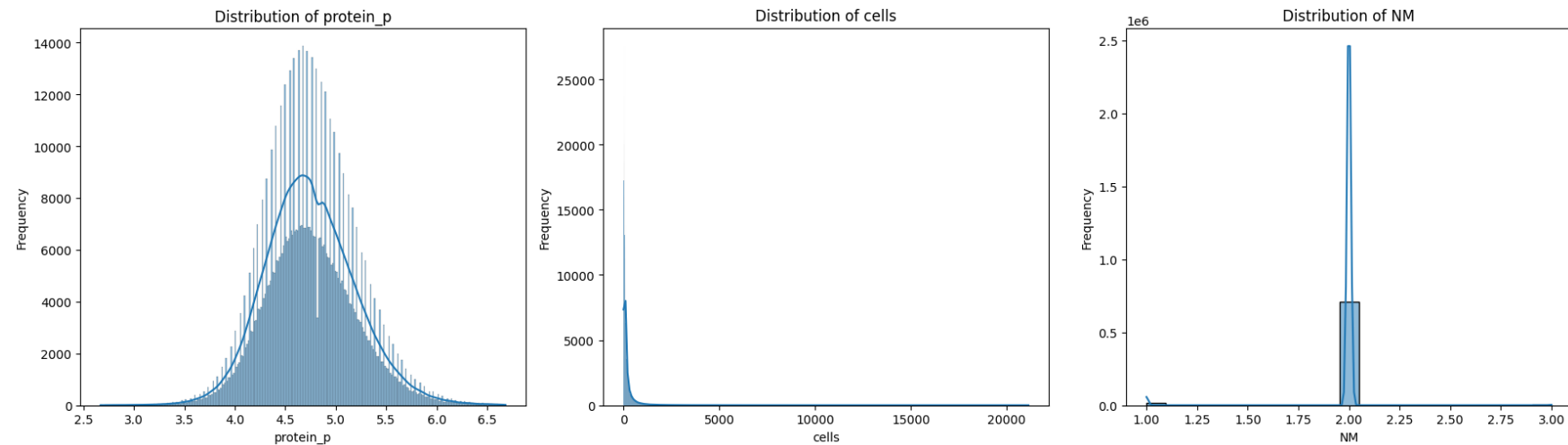


Dataset Overview & Base Variables

Base Data



After Removing outliers



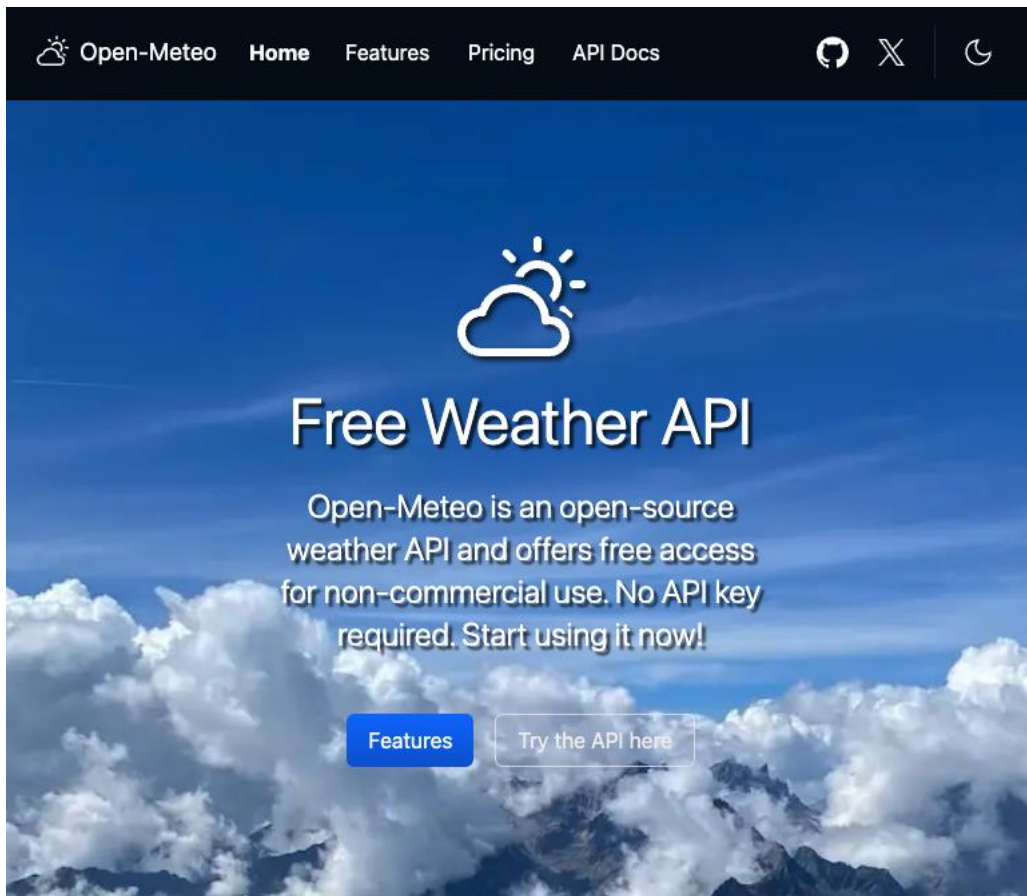
Weather Data



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Weather Data



Open-Meteo weather API

Period : Hourly

Start Date : 2013-01-01

End Date : 2025-12-31

Location : 329 individual locations

Variables :

- Temperature_2m
- relative_humidity_2m
- wind_speed_10m

Merged Data



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Merged Data

2,593,267 rows

33 columns:

- Farm_Code
- Animal_ID
- dtb
- dtc
- dtt
- parity
- milk_kg
- fat_p
- protein_p
- cells
- NM
- ECM
- SCS
- AFC
- DIM
- temperature_avg
- temperature_max
- temperature_min
- humidity_avg
- humidity_max
- humidity_min
- wind_speed_avg
- wind_speed_max
- wind_speed_min
- THI_1_avg
- THI_1_min
- THI_1_max
- THI_2_avg
- THI_2_min
- THI_2_max
- WHI_avg
- WHI_min
- WHI_max

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